

01/12/2023

Time series analysis of S&P Green Bond Index



The IFoA's Data Science, Sustainability and Climate Change Working Party shares its initial time series analysis of the S&P Green Bond Index using non-traditional techniques including neural networks.

Introduction

On behalf of the IFoA's Data Science, Sustainability and Climate Change Working Party, we are pleased to publish our first paper as a working party: [Time series analysis of GSS bonds: Part 1](#)

The paper focusses on the initial stages of our analysis where we try to replicate the S&P Green Bond Index using non-traditional techniques including neural networks such as CNNs, LSTMs, and GRUs. We extend our analysis to include a popular decision tree model called XGBoost.

The paper gives details of the underlying model architecture, as well as

Please note that the paper focusses on the initial stages of our analysis and acts as the foundation for future publications. We have therefore deliberately excluded discussing topics such as stationarity and correlation with the general market, which are deferred for future papers.

The code used Python-based libraries including Keras and Google's Tensorflow framework. The environment used for our analysis was Google Colab. For those interested, please see for example Google's introductory tutorials via their website for Tensorflow (see link below), which also include tutorials on time series analysis with neural networks. We have used the open-source library Optuna to assist with hyperparameter optimisation. These are discussed further in the paper.

Background to working party

The working party, formed in 2022, is looking at a range of areas within the data science, sustainability, and climate change space. We aim to add value in addressing sustainability and climate change issues with data science techniques, as well as using this opportunity to educate the wider actuarial profession in these areas.

Do keep an eye out for [future publications by our working party](#).

Further reading

- [IBM's introduction to data science](#)
- [IFoA's data science research website](#)
- [IFoA's lifelong learning data science page](#)
- [Pytorch](#)
- [Tensorflow](#)
- [IFoA's Data Science Community](#)

Share your views

What are your thoughts on the points raised in this article and [accompanying paper](#)? We would love to hear your views in the comments via the [IFoA's](#)



Authors



Debashish Dey

Co-chair, Climate Change Disclosures Working Party and Data Science, Sustainability and Climate Change Working Party; Chair, Unsupervised Learning Working Party until 6 May 2025



IFoA's Data Science, Sustainability and Climate Change Working Party

Popular topics

[AI](#)[Actuarial reports](#)[Annuities](#)[Banking](#)[Biodiversity](#)[CACT](#)[CMI](#)[Careers](#)[Chartered Actuary](#)[Climate Change](#)[Covid-19](#)[DEI](#)[IFoA Asia Conference](#)[IFoA Foundation](#)[IFoA conferences](#)[Neurodiversity](#)[Pensions](#)

All topics

[Contact Us](#)[Site map](#)[Visit Us](#)[Cookie Policy](#)[Accessibility Statement](#)[Working at the IFoA](#)[Terms and Conditions](#)[Find an Actuary](#)[The Actuary](#)



[The Actuary Jobs Asia](#)

[Privacy notice](#)

[Refund policy](#)

[Shop](#)

[CMI](#)



Copyright © Institute and Faculty of
Actuaries, 2014-2025.

